

Derivatives of Exponential or Logarithmic Functions and More Practice with Differentiation (Homework)
Differentiate.

① $f(x) = \left(\frac{-x+5}{3x+1}\right)^6$

② $y = \sqrt{\frac{x^2}{4x^3+6}}$

$$\textcircled{3} f(x) = 5x^3 \sin x$$

$$\textcircled{4} y = -4\sqrt[5]{x+3} \csc x$$

Simplify each function. Then, find the derivative.

⑤ $g(x) = 2x^5(3x^3 + 4x - 1)$

⑥ $h(x) = \frac{5x^3 + 3x^2 + 1}{2x}$

⑦ $f(x) = \frac{\sqrt[4]{x^6} + 3}{x}$

⑧ $g(x) = x^2(\sqrt{x} - 7)$

Find $\frac{dy}{dx}$.

⑨ $y = 6^x$

⑩ $y = 11^x$

$$\textcircled{11} y = 7^{\cot x}$$

$$\textcircled{12} y = 8^{\sqrt{x}}$$

Differentiate.

$$\textcircled{13} y = e^x$$

$$\textcircled{14} y = e^{x^3 - 2x - 4}$$

$$\textcircled{15} y = e^{\cos x + 6}$$

$$\textcircled{16} f(x) = \log_6 x$$

$$\textcircled{17} g(x) = \log_7 x$$

$$\textcircled{18} h(x) = \log_2 \left(\frac{6}{x^4} \right)$$

$$\textcircled{19} \quad y = \log_{15}(\sec x)$$

$$\textcircled{20} \quad f(x) = \ln x$$

$$\textcircled{23} \quad f(x) = \ln(\cot x)$$

$$\textcircled{24} \quad f(x) = \ln\left(\frac{1}{x}\right)$$

$$\textcircled{21} \quad f(x) = \ln(1-x^2)$$

$$\textcircled{22} \quad f(x) = \ln(6x^4)$$

$$\textcircled{25} \quad f(x) = \ln(-10x^2 + \sin x)$$

Homework Answers

$$\textcircled{1} \quad -\frac{96(5-x)^5}{(3x+1)^7}$$

$$\textcircled{2} \quad \frac{-2x^3+6}{\sqrt{(4x^3+6)^3}} \text{ or } -\frac{x^3-3}{\sqrt{2(2x^3+3)^3}}$$

$$\textcircled{3} \quad 15x^2 \sin x + 5x^3 \cos x$$

$$\textcircled{4} \quad \frac{20(x+3) \csc x \cot x - 4 \csc x}{5 \sqrt[5]{(x+3)^4}}$$

$$\textcircled{5} \quad 48x^7 + 48x^5 - 10x^4$$

$$\textcircled{6} \quad 5x + \frac{3}{2} - \frac{1}{2x^2}$$

$$\textcircled{7} \quad \frac{1}{2\sqrt{x}} - \frac{3}{x^2}$$

$$\textcircled{8} \quad \frac{5}{2} \sqrt{x^3} - 14x$$

$$\textcircled{9} \quad 6^x \ln 6$$

$$\textcircled{10} \quad 11^x \ln 11$$

$$\textcircled{11} \quad -7^{\cot x} (\ln 7) \csc^2 x$$

$$\textcircled{12} \quad \frac{8^{\sqrt{x}} \ln 8}{2\sqrt{x}}$$

$$\textcircled{13} \quad e^x$$

$$\textcircled{14} \quad (3x^2 - 2)e^{x^3 - 2x - 4}$$

$$\textcircled{15} \quad -e^{\cos x + 6} \sin x$$

$$\textcircled{16} \quad \frac{1}{x \ln 6}$$

$$\textcircled{17} \quad \frac{1}{x \ln 7}$$

$$\textcircled{18} \quad -\frac{4}{x \ln 2}$$

$$\textcircled{19} \quad \frac{\tan x}{\ln 15}$$

$$\textcircled{20} \quad \frac{1}{x}$$

$$\textcircled{21} \quad -\frac{2x}{1-x^2}$$

$$\textcircled{22} \quad \frac{4}{x}$$

$$\textcircled{23} \quad -\frac{\csc^2 x}{\cot x} = -\csc^2 x \frac{1}{\cot x} = -\csc^2 x \tan x$$

$$= -\frac{1}{\sin^2 x} \frac{\sin x}{\cos x}$$

$$= -\frac{1}{\sin x} \frac{1}{\cos x}$$

$$= -\csc x \sec x$$

$$\textcircled{24} \quad -\frac{1}{x}$$

$$\textcircled{25} \quad \frac{-20x + \cos x}{-10x^2 + \sin x}$$